

DATABASE SYSTEMS PROJECT

Disaster Relief Volunteer Registry

Skills-based matching for emergency deployment

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Code: <https://github.com/Allam-dev/dbproject>

Project Overview



Problem

Coordinate disaster relief volunteers by matching their skills to emergency event requirements in real time.



Goal

Build a normalized PostgreSQL database with a GUI to manage volunteers, skills, and disaster events.

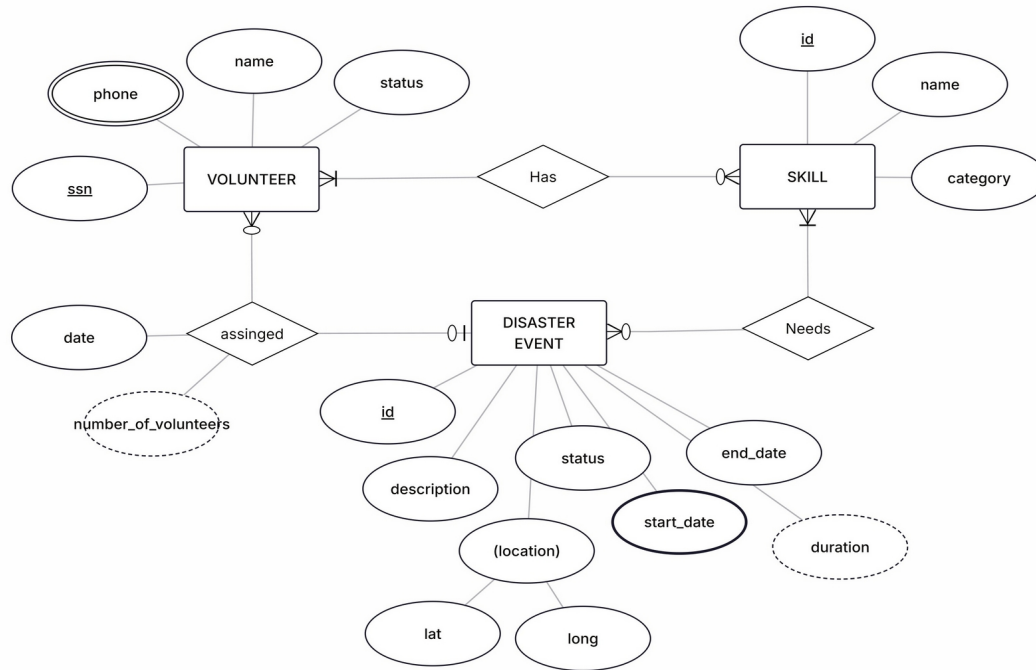


Solution

ER model → Relational schema → DDL + sample data → Flutter desktop app connected to PostgreSQL.

Tech Stack: PostgreSQL • Flutter (Desktop) • SQL DDL • ER Modeling

Entity-Relationship Diagram



3 Entities • 3 Relationships • Multivalued & Derived Attributes

ERD — Entities & Relationships

VOLUNTEER

PK: SSN · Attributes: name, status · Multivalued: phone ·
Derived: number_of_volunteers

DISASTER EVENT

PK: id · Attributes: description, status, start_date, end_date ·
Composite: location (lat, long) · Derived: duration

SKILL

PK: id · Attributes: name, category

Has (VOLUNTEER ↔ SKILL)

M:N — A volunteer can have many skills; a skill can belong to many volunteers

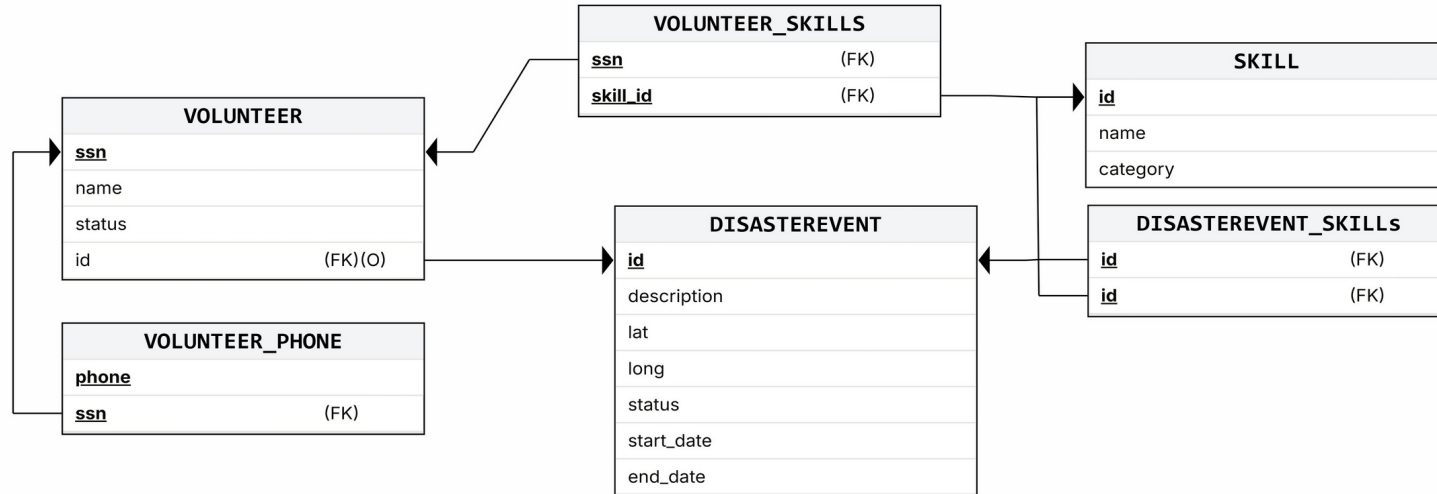
Needs (EVENT ↔ SKILL)

M:N — An event can require many skills; a skill can be needed by many events

Assigned (VOLUNTEER ↔ EVENT)

1:N — A volunteer is assigned to at most one active event; an event has many volunteers

Relational Schema



6 tables · Foreign key constraints · ON DELETE CASCADE & SET NULL policies

Schema — Table Summary

Table	Primary Key	Foreign Key(s)	Notes
SKILL	id (SERIAL)	—	Stores skill name & category
DISASTEREVENT	id (SERIAL)	—	Stores event details, location, dates, status
VOLUNTEER	ssn (VARCHAR)	id → DISASTEREVENT	Optional FK for current assignment
VOLUNTEER_PHONE	(phone, ssn)	ssn → VOLUNTEER	Multivalued phone attribute
VOLUNTEER_SKILLS	(ssn, skill_id)	ssn, skill_id	M:N junction table
DISASTEREVENT_SKILLS	(event_id, skill_id)	event_id, skill_id	M:N junction table

DDL — Key Definitions

VOLUNTEER Table

```
CREATE TABLE VOLUNTEER (  
  ssn    VARCHAR(20) PRIMARY KEY,  
  name   VARCHAR(100) NOT NULL,  
  status VARCHAR(50) DEFAULT 'available',  
  id     INT REFERENCES DISASTEREVENT(id)  
        ON DELETE SET NULL  
);
```

VOLUNTEER_SKILLS Junction

```
CREATE TABLE VOLUNTEER_SKILLS (  
  ssn      VARCHAR(20)  
    REFERENCES VOLUNTEER(ssn)  
    ON DELETE CASCADE,  
  skill_id INT  
    REFERENCES SKILL(id)  
    ON DELETE CASCADE,  
  PRIMARY KEY (ssn, skill_id)  
);
```

Design Decisions

- **SSN as PK:** Natural, globally unique identifier for volunteers avoids surrogate key overhead
- **ON DELETE SET NULL:** Volunteer's event assignment becomes NULL when an event is deleted — preserves volunteer record
- **ON DELETE CASCADE:** Junction tables auto-clean when a volunteer or skill is deleted
- **SERIAL PKs:** Auto-incrementing IDs for SKILL and DISASTEREVENT for simplicity and performance

Sample Data

Volunteers

SSN	Name	Status
123-45-6789	Ahmed Hassan	Deployed
987-65-4321	Sara Mohamed	Available
111-22-3333	Omar Khalil	Available

Disaster Events

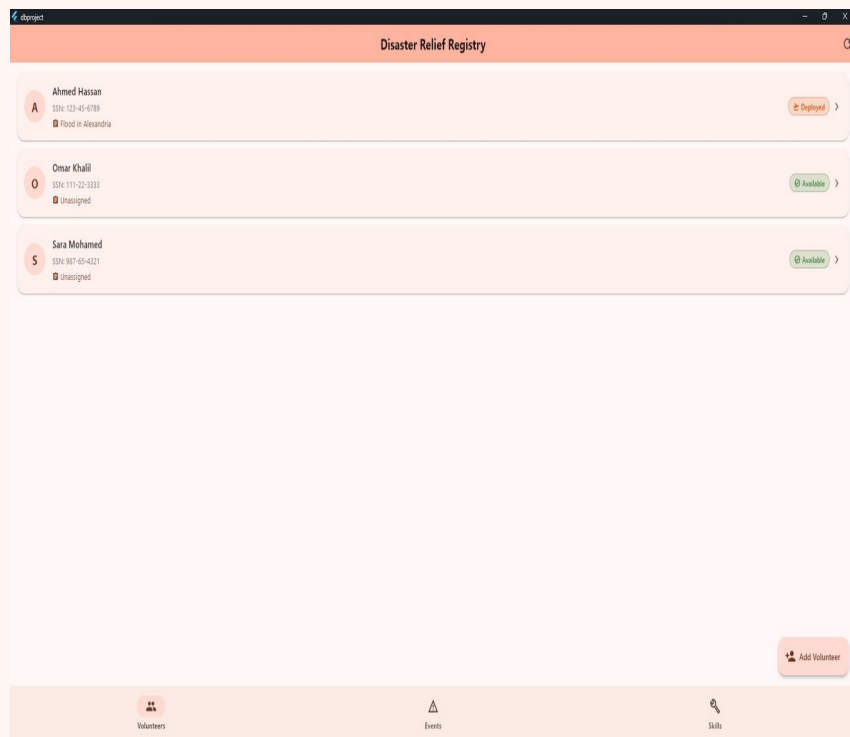
Description	Status	Start Date	Required Skills
Flood in Alexandria	Active	Jan 10, 2025	Nursing, CDL License
Earthquake response	Resolved	Nov 1, 2024	Search & Rescue

Skills

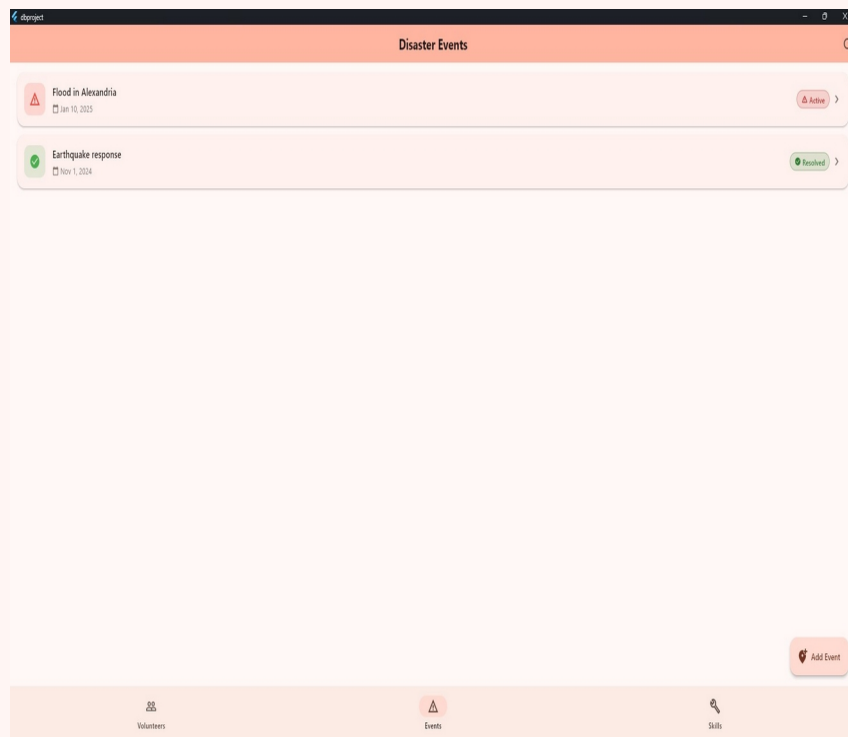
Name	Category
Nursing	Medical
CDL License	Transport
First Aid	Medical
Search & Rescue	Emergency
Arabic	Language

Application GUI — Volunteers & Events

Volunteer Registry



Disaster Events



Built with Flutter · Connected to PostgreSQL · Bottom navigation: Volunteers | Events | Skills

Application GUI — Skills & Add Event

Skills (grouped by category)

Skills

Emergency

- Search & Rescue
Emergency

Language

- Arabic
Language

Medical

- First Aid
Medical
- Nursing
Medical

Transport

- CDL License
Transport

+ Add Skill

Volunteers Events Skills

Add Event Form

Add Event

Description

Latitude Longitude

Status
Active

Select Start Date
Start Date *

Select End Date (Optional)
End Date

Required Skills

Search & Rescue (Emergency) Arabic (Language) First Aid (Medical) Nursing (Medical) CDL License (Transport)

Create Event

Skills displayed by category · Event form with skill chip selection · Add Skill modal

Summary & Deliverables

What We Built



ER Diagram

3 entities, 3 relationships, derived & multivalued attrs



Relational Schema

6 normalized tables with FK constraints



PostgreSQL DDL

Full CREATE TABLE statements + sample data



Flutter GUI

Desktop app: Volunteers, Events, Skills screens